

Business Question and Visualization Report

Date	4 October 2026
Team ID	N/A
Project Name	Student Mental Health Analysis (Tableau Dashboards & Story)
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting.

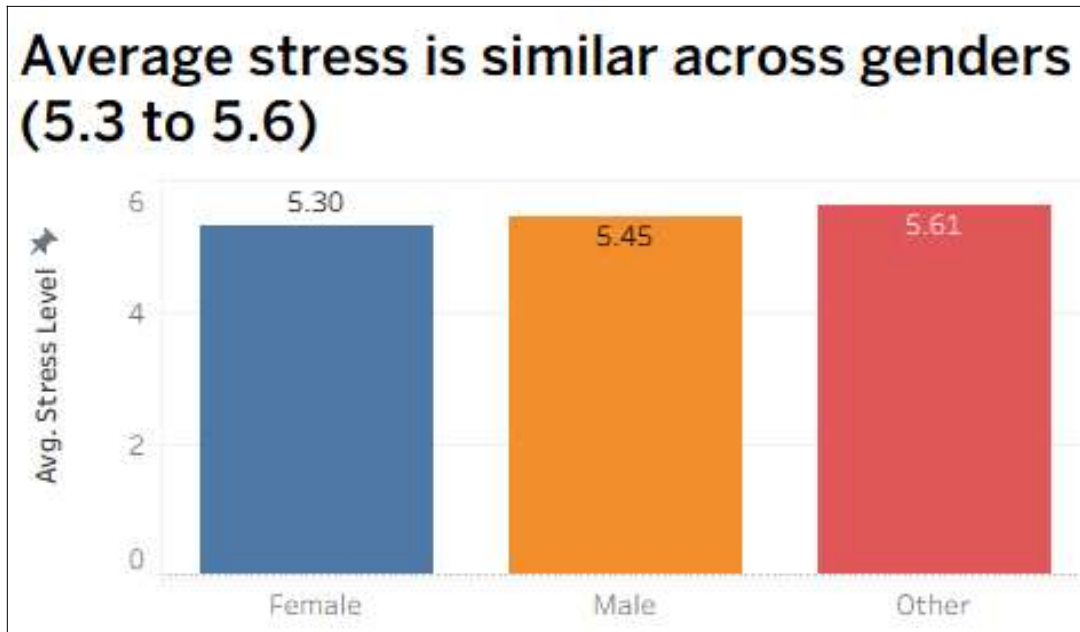
Business Questions and Visualisation: The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. What are the overall wellbeing and attendance levels of the students? *Visualization:* KPI tiles showing the average mental score (5.45), sleep duration (6.49 hours), depression level (5.50) and attendance rate (79.96%), with a Gender filter that updates every chart.

Mental Score	Sleep Duration Hours	Depression Level	Attendance Rate (%)
5.45	6.49	5.50	79.96

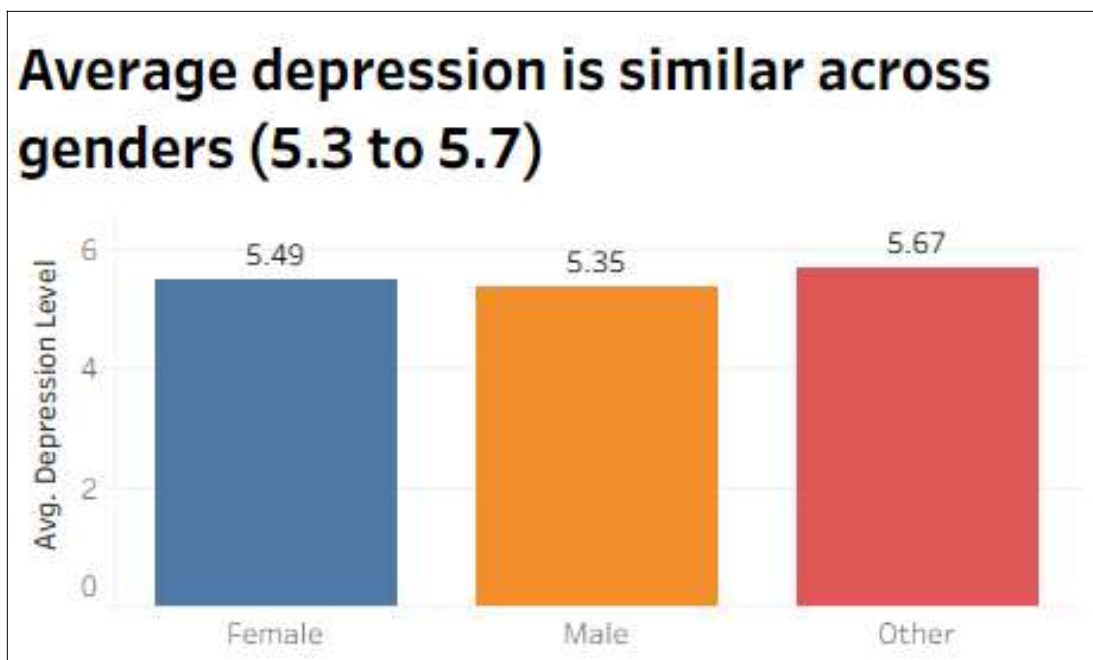
2. How does the average stress level differ between genders?

Visualization: Bar chart of average stress level by gender: Female 5.30, Male 5.45, Other 5.61. The gap is very small.



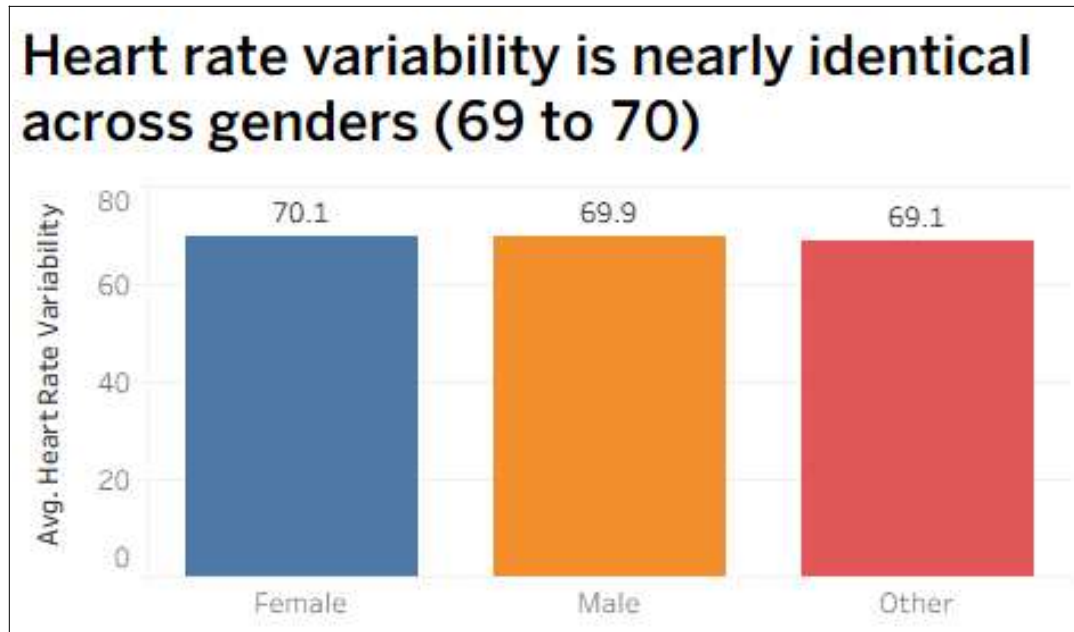
3. How does the depression level vary across genders?

Visualization: Bar chart of average depression level by gender: Female 5.49, Male 5.35, Other 5.67. Again a very small gap.



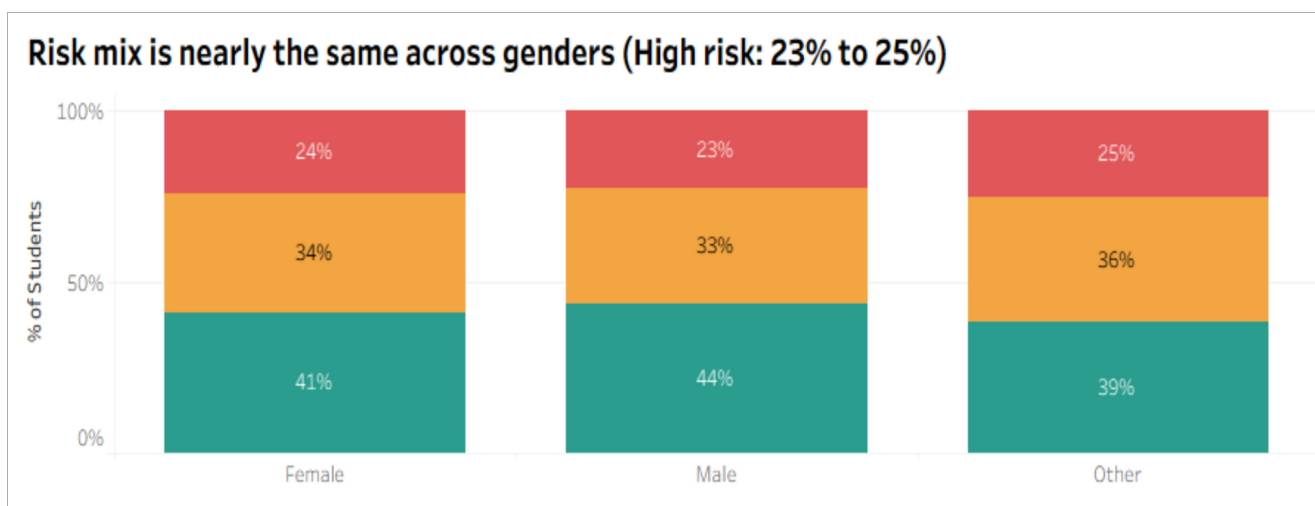
4. Does heart rate variability differ between genders?

Visualization: Bar chart of average heart rate variability by gender: Female 70.1, Male 69.9, Other 69.1. Nearly identical.



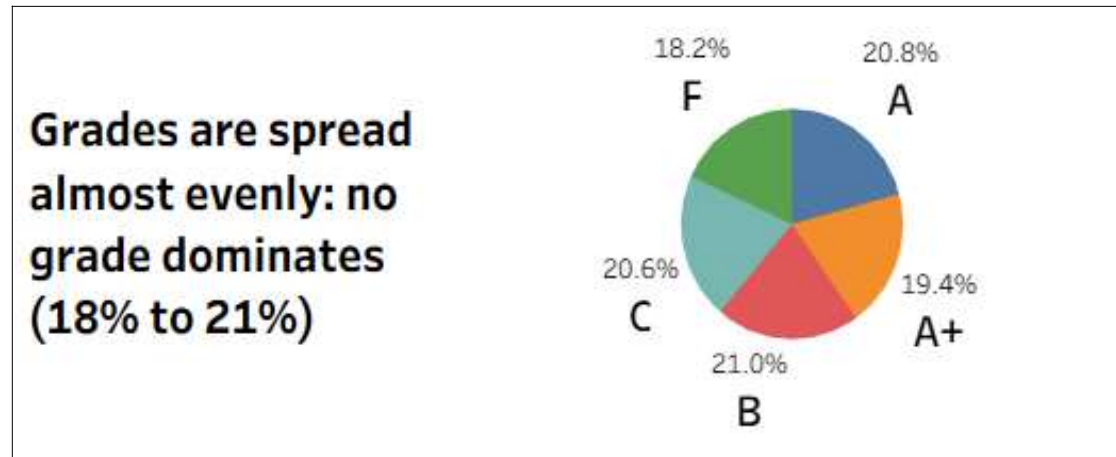
5. How is the mental health risk level distributed across genders?

Visualization: 100% stacked bar chart of Low, Medium and High risk by gender. High risk is 24% for Female, 23% for Male and 25% for Other, so risk is spread evenly.



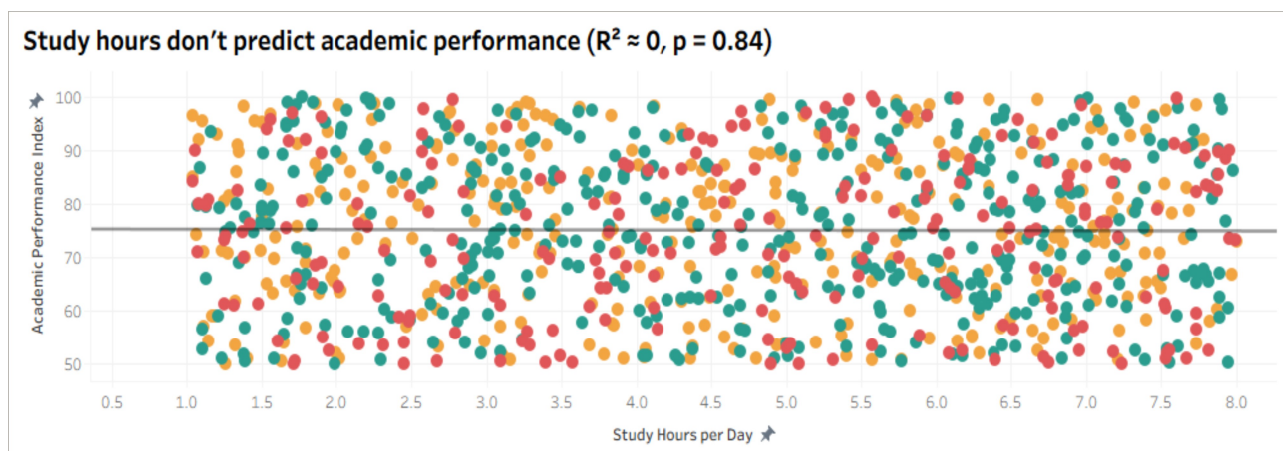
6. How are students spread across academic grades?

Visualization: Pie chart of the grade split: A 20.8%, A+ 19.4%, B 21.0%, C 20.6% and F 18.2%.
No grade dominates.

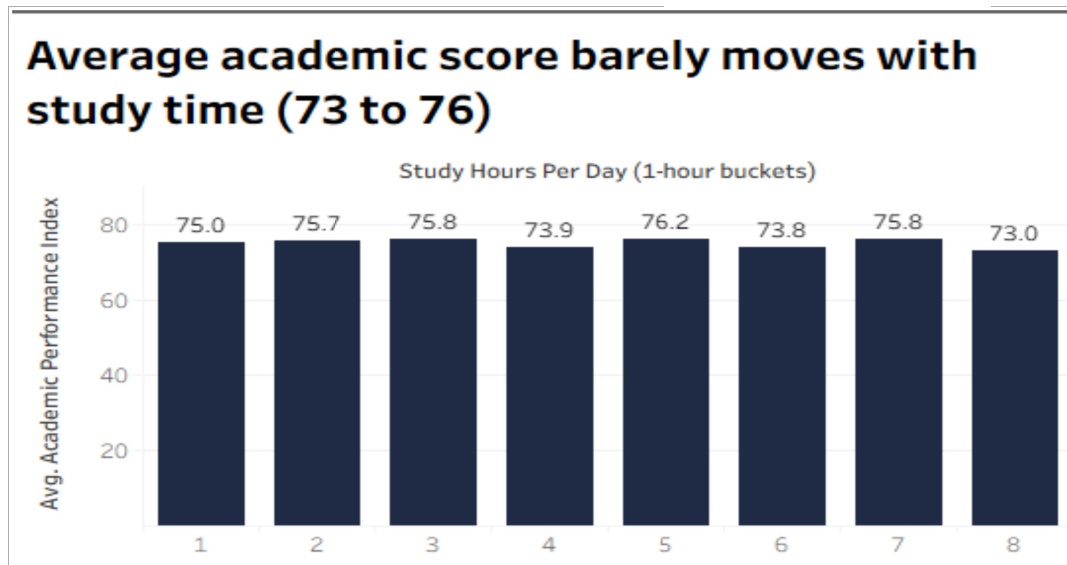


7. Do more study hours lead to better academic performance?

Visualization: Scatter plot of study hours per day against the Academic Performance Index with a trend line. The line is flat ($R^2 \approx 0$, $p = 0.84$), so study hours do not predict academic score.



8. How does the average academic score change with study time?



Visualization: Bar chart that groups study hours into 1-hour buckets and shows the average academic score for each. The score stays between 73 and 76 in every bucket.